

Axisymmetric Navier-Stokes with swirl: the $1/r^4$ axis term (without Φ -renormalization)

Status: OPEN obstruction. Not a Clay proof. Not ChatVault. This is the operator before the algebraic cancel.

Hunt note. Zenodo records 22050974 / 22050975 / 22045467 / 21071991 / 20405405 / 20405597 all introduce Φ -renormalization. No separate public “without cancel” PDF was found. Domain Architect therefore states the pre-cancel operator here and points at the later live face.

Operator (still in the equations). Axisymmetric-with-swirl, cylindrical (r, θ, z) . Angular momentum $\Gamma = r u_\theta$. Renormalized azimuthal vorticity $\Omega = \omega_\theta/r$. The swirl-vorticity source still carries the centrifugal axis term:

$$D_t \Omega = (1/r^4) \partial_z(\Gamma^2) + \nu L_{\text{cyl}} \Omega$$

with $D_t = \partial_t + u_r \partial_r + u_z \partial_z$. As $r \rightarrow 0$ the $1/r^4$ prefactor is the classical axis obstruction. Standard H^1 estimates meet a Hardy inequality with weight r^{-4} .

Cite R. Danchin (2007). Live Φ note 10.5281/zenodo.22050974 §1.2 [6] says the $1/r^4$ term is explicitly identified in Danchin, Axisymmetric incompressible flows with bounded vorticity, Russian Math. Surveys 62 (2007), 73–94, as the obstruction to direct energy methods. That is a diagnosis of this WITHOUT-cancel operator. It does not cancel the term, does not prove swirl regularity, and does not close unaugmented 3D Navier-Stokes.

Also named on that live PDF, and also not a fill. Ladyzhenskaya (1968) and Ukhovskii-Yudovich (1968): without swirl, regularity is classical — does not apply to with-swirl large data. Chae-Lee (2002): small swirl. Chen-Fang-Zhang (2017): regularity if swirl is in $L_t^\infty L_x^3$ (criterion, not a bound). Hou-Li (2008): dynamic stability / blow-up analysis with boundary.

Gap WITHOUT $\rightarrow$ WITH (UNFILLED). Matching the live WITH-cancel face requires applying $\Phi = \Gamma/r^2 = u_\theta/r$ so that $r^{-4} \partial_z(\Gamma^2) = \partial_z(\Phi^2)$. Danchin names the hole; the identity is not Danchin’s theorem. Next attempt (not RH/Goldbach): Chen-Fang-Zhang $L_t^\infty L_x^3$ swirl criterion — still open because the bound is not in hand on this operator.

What this face does not do. It does not apply $\Phi = \Gamma/r^2$. It does not use $r^{-4} \partial_z(\Gamma^2) = \partial_z(\Phi^2)$. It does not claim global regularity, Clay Statement B, or RH.

Later face (WITH cancel). Live Φ -renormalization (Q1-augmented / Φ system): DOI 10.5281/zenodo.22050974, file 01_phi_renormalization.pdf. Sibling 10.5281/zenodo.22050975. June conditional Φ : 10.5281/zenodo.21071991 (archive). That later face still does not claim unconditional classical 3D Navier-Stokes: Q1 $\neq$ classical. LPS bootstrap there is for the augmented PDE.

Domain Architect is a Functional Role Analysis classifier, not a proof engine. Canonical SFE unresolved. Unconditional NS smoothness is a hypothesized realization, not a theorem.